

RESEARCH PROJECT 03: LITERATURE REVIEW & COMPARISON

TEAM MEMBERS: 25I-7636 & 25I-7601

LITERATURE REVIEW

Paper 01: A Novel Hybrid Approach for Driver Drowsiness Detection Using a Custom Deep Learning Model

This paper addresses the challenge of real-time detection of driver drowsiness in complex driving conditions, which aim is to create a robust automatic system that can detect drowsiness based on behavioral characteristics through video surveillance. The authors present a comprehensive model titled D3-CDLM (Driver Drowsiness Detection using Custom Deep Learning Model), which has four modules: feature extraction and selection, machine learning and ensemble techniques, deep learning, and hybrid combination. The study utilizes the **Yawning Detection dataset**, which contains 322 videos from the front mirror camera and 29 videos from the dashboard camera with 30 frames per second at 640x480 resolution. The dataset has three classes, normal/not yawning (21,400 samples), yawning (14,272 samples), and talking (19,925 samples).

The **proposed method** is based on the use of the HOG feature extraction method. The HOG method is rotation and illumination invariant. PCA is applied to reduce the dimensionality of the features to select the optimal HOG features. The proposed 30-layer CNN architecture is based on the use of the hybrid module that combines PCA-controlled HOG features with the proposed CDLM. For the proposed machine learning method, they compared the performance of SVM, Decision Tree, Linear Regression, Random Forest, and XGBoost.

The **experimental results** show that the authors achieved 99.4% accuracy for ternary classification and 99.7% accuracy for binary classification with 60 epochs. The proposed hybrid method achieved 99.6% accuracy. The proposed method also achieved 97.4% accuracy with the use of the XGBoost algorithm with the hybrid method. The proposed method achieved better performance compared to the existing state-of-the-art methods including CNN (11.8%), 6TDBH-HMM (14.0%), and Yolov3 (4.5%). However, the study has several **limitations**, as it only considers yawning as a sign of drowsiness, which might miss other signs of fatigue. The data set might not contain enough data about driving at night. Although the 30-layer architecture is effective, degradation could be a problem if the number of layers increases. Individual differences in the manifestation of drowsiness in drivers are not considered in the study.

Paper 02: Research on a Real-Time Driver Fatigue Detection Algorithm Based on Facial Video Sequences

The objective of the study is to increase the accuracy of the detection of driver fatigue using a comprehensive real-time algorithm based on facial landmarks. The study tries to solve the problem of the current detection system based only on the detection of a single feature, which does not provide adequate accuracy and often results in false recognition. The research is based on the **FDDb dataset**, consisting of 5171 faces from wild data to validate face detection. A data set collected from 10 drivers (20 to 45 years old, 5 females) who conducted 30 driving tests, each consisting of a 30-minute drive in a simulated two-lane motorway.

The system is designed using a Tasks-Constrained Deep Convolutional Network (TDCDN) model for face region detection with 68 key points, which solves optimization problems depending on the different speeds of task convergence. The algorithm also computes the Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and PERCLOS (percentage of eye closure time). A KNN classifier with the computed values of EAR, PERCLOS, and MAR is used. A comprehensive fatigue assessment model with weighted features (EAR: 0.2, PERCLOS: 0.8, MAR: 0.1) is used for determining the fatigue level.

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The results indicate that the proposed algorithm obtained an accuracy level of 95.1%. The EAR value for the proposed algorithm was set at 0.2, while the PERCLOS value was set at 0.8 (p80). The MAR value for the proposed algorithm was set at 0.6. The system also successfully classifies the level of fatigue, ranging from mild ($10 < N \leq 20$), moderate ($20 < N < 50$), and severe ($N \geq 50$), depending on the frames. The proposed adaptive method has shown different sensitivity (44-64%) with different videos due to camera position changes and the combination of video segments. The facial landmark detector of the Dlib library also has some **limitations** with extreme camera angles, insufficient lighting conditions, and small face sizes. The linear method is not effective to distinguish the fatigue state from other emotional states.

Paper 03: Drowsiness Detection in Drivers Using Facial Feature Analysis

The problem addressed in this research is the necessity to improve the accuracy of drowsiness detection, particularly for long-haul travel. In this study, two algorithms, static and adaptive frame thresholds, are proposed to improve road safety via non-invasive techniques. In this study, two datasets, **UTA-RLDD** and **YawDD**, are used. UTA-RLDD comprises 180 videos, or 30 hours of RGB video, from 60 participants. YawDD contains 254 videos with various yawning scenarios. Videos are cut into four segments, each of 30 minutes, with a frame rate of 15 fps and a resolution of 320x240 pixels.

The **methodology** includes, Dlib's facial feature detection algorithm, a 68-point facial landmark detector, is used to calculate the Eye Closure Ratio and Mouth Aperture Ratio. In the static threshold algorithm, thresholds are fixed for both ECR and MAR, i.e., 0.15 and 0.4, respectively. In the adaptive frame threshold algorithm, the count of consecutive frames changes based on the driving time, i.e., for MAR, it changes from 90 to 60, and for ECR, it changes from 60 to 8.

The static threshold method showed 89.4% accuracy and 96.5% sensitivity by using 1000 images. The adaptive frame threshold method showed up to 98.2% accuracy and 64.3% sensitivity by using four videos of 30 minutes duration. The system is cost-effective as it does not require any training and can process in real-time. It was **observed** that the sensitivity obtained by the adaptive method was very low, ranging from 44-64%, depending on the videos, as the camera position and video segments were merged. The facial landmark detector provided by Dlib has some limitations in terms of extreme angles, lighting, and face sizes. The accuracy obtained by the linear method may not be effective in differentiating between fatigue and emotion.

Paper 04: Detection of Drivers Drowsiness Using Adaptive Eye Characteristic Ratio for Enhanced Road Safety

In this paper, a real-time system for the detection of drivers' sleeps and timely alert generation via eye detection is proposed. The problem of fixed thresholds is solved by incorporating adaptive thresholds based on individual drivers' eye characteristics. In this study, the **NTHU Drowsy Driver Detection Dataset**, comprising 36 subjects and five scenarios, i.e., glasses, without glasses, glasses at night, without glasses at night, and sunglasses, with a total of 722,223 images for training, 173,259 images for evaluation, and 736,132 images for testing, is used.

The **implementation** involves, Dlib's facial landmarks detector, i.e., HOG+Linear SVM, is used for face and eye detection. Adaptive thresholds are calculated by taking the average of the top and bottom 50 Eye Characteristic Ratio values obtained from a 30-second window, i.e., 15 seconds with eyes closed and 15 seconds with eyes open. In this study, four classifiers, i.e., Random Forest, Decision Tree, KNN, and Multinomial, are used for classification.

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In the experiment, the Random Forest classifier outperformed with an accuracy of 90.48%, followed by Decision Tree, KNN, and Multinomial, with accuracies of 89.44%, 88.63%, and 84.32%, respectively. The accuracy of blink detection was found to be 99.784%. AUC for each classifier was obtained as follows, Random Forest, 0.9088; Decision Tree, 0.8983; KNN, 0.8844; and Multinomial, 0.8464. The adaptive threshold effectively takes care of individual differences in the shapes of the eyes. **However**, the focus of this research is only on eye features, not including mouth or head position indicators. The calibration step, which needs cooperation from the driver, might not be feasible in a real-world application. The performance under extreme lighting conditions needs to be validated. The subjects of the dataset are only 36 people.

Paper 05: DrowsyDetectNet: Driver Drowsiness Detection Using Lightweight CNN With Limited Training Data

This research solves the problem of building lightweight drowsiness detection models, which can be deployed in environments with scarce resources, as in vehicle embedded systems. The research presents DrowsyDetectNet, a lightweight CNN model for effective operation with scarce training data while ensuring a high level of accuracy. They make use of two datasets: Dataset-1, which contains 324 images from a previously published research paper by Chirra et al., and Dataset-2, which contains 1,452 images from the "**yawn_eye_dataset_new**" available on the Kaggle platform.

The **proposed system** makes use of a 68-point facial landmark detector (Dlib) for detecting the face region and the eye region. The shallow CNN has four convolutional layers, four max pooling layers, two dropout layers, and two fully connected layers. The hyperparameter optimization technique in which the learning rate ranges from 0.1 to 0.0001, the number of epochs ranges from 10 to 100, the batch size ranges from 4 to 32.

The proposed shallow CNN reported 99.23% accuracy on Dataset-1 and 99.14% accuracy on Dataset-2 with 100 epochs, batch size 32, learning rate 0.001, and Adam optimizer. InceptionV3 (96.15%, 93.12%), MobileNetV2 (97.69%, 91.06%), ResNet50 (98.46%, 98.1%), VGG19 (98.46%, 98.62%). Inference time of shallow CNN was significantly less (1430ms) compared to InceptionV3 (3233ms) for Dataset-1. The dataset **lacks** diversity in illumination conditions, ethnicity, and head pose. The performance of the proposed system in low illumination conditions, especially at night, should be verified with infrared and near-infrared images.

Paper 06: Facial Feature-Based Drowsiness Detection with Multi-Scale Convolutional Neural Network

The paper introduces a Multi-Scale Convolutional Neural Network framework for drowsiness classification to avoid road accidents resulting from driver fatigue. The study attempts to address the requirement for precise detection systems via the incorporation of optimization techniques and deep learning models. The research employs two datasets for experimentation, namely, **YAWDD**, yawning detection with in-car cameras, and **NTHU-DDD**, consisting of 36 subjects with varying scenarios and with/without glasses and night conditions, totaling 9.5 hours. Data distribution: 80% for training and 20% for testing. In YAWDD, there are a total of 10,314 images, out of which 5,594 images are for the drowsy class and 3,639 images for the non-drowsy class.

The methodology includes, for key-frame landmark detection, the Dlib library was used. Noise was removed by using a Cross-Guided Bilateral filtering algorithm for preprocessing followed by feature vector extraction through using the hybrid Dual-Tree Complex Wavelet Transform / Walsh-Hadamard Transform.

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Feature selection optimized through a Flamingo Search Algorithm modeling foraging and migration behaviors. The classification process used to classify the output used MCNN; convolutional neural network; max pooling; and classification.

The new MCNN with FSA had an accuracy of 98.38% on the YAWDD dataset and 98.26% on the NTHU-DDD dataset. The performance comparison of the proposed MCNN on the YAWDD dataset showed substantial accuracy improvements over other architectures, including AlexNet (95.3%), ResNet (95.5%), VGGNet (95.9%), SVM (93.33%), and Random Forest (90.34%) as compared with the proposed MCNN (98.38%). Likewise, the performance results for the NTHU-DDD dataset were also excellent at an accuracy rate of 98.26%. **However**, Real-time implementation may not be possible because of the computation complexity of the study. The optimization method requires additional processing. There is not enough detail on diversity of samples across ethnicities and its effect on age groups. The performance of the algorithm under extreme occlusion needs to be validated more thoroughly. Errors can accumulate across the multi-stage pipeline.

Paper 07: Real-Time Machine Learning-Based Driver Drowsiness Detection Using Visual Features

This paper describes a non-intrusive way to monitor drowsiness in real time by analyzing visual information taken from dashboard-mounted cameras. It provides a thorough method for monitoring drowsiness by combining head pose estimation, facial landmarks and facial mesh detection with several classifiers. To carry out the research, they collected data for 23 individuals - 18 for training and 5 for testing - from various behavioral states: distracted, engaged in conversation, laughing, showing sleepiness with their eyes closed, yawning, and/or nodding. Videos were captured (at a resolution of 640 x 480 at 30 frames per second) under a variety of conditions: with or without glasses, as well as at night.

In the pre-processing of data, the Dlib HOG was used to detect the individual's eyes and mouth, while MediaPipe was used to estimate the individual's 3D head pose. The following features were extracted from the video data: Eye Aspect Ratio (EAR) - calculated from six (6) eye landmarks; Mouth Aspect Ratio (MAR) - calculated from eight (8) mouths; and head pose angles (X/Y rotation) with thresholds of ± 7 degrees). Drowsy events were monitored with a 15-frame temporal window (the frames had to be from within 500 milliseconds of each other). Each classifier was evaluated: Random Forest - consisting of 50 decision trees trained using the entropy criterion; Sequential Neural Network - a six-layer neural network employing ReLU/sigmoid activation functions; and Linear SVM - where $C=1$.

The Random Forest classifier had an accuracy rate of 99%, a sensitivity rate of 99%, a specificity rate of 98% and an AUC score of 0.999. The Sequential Neural Network resulted in an accuracy rate of 96% and the Linear SVM produced an accuracy rate of 80%. The system's initial decision requires 1 second (30 frames) to make and all future decisions are made every 33ms. Systems were developed to determine a baseline EAR threshold of 0.2 and a baseline MAR threshold of 0.5. However, the **limitations** of the study include, HOG face detector will be inaccurate due to the lighting level, occlusions, and/or the face being in motion; the assumption of 15 frames (500ms) per individual blink time may not be applicable; head pose threshold of ± 7 degrees may not work for all drivers; and the lack of validation against a real-world driving environment as opposed to a simulation environment.

Paper 08: Novel Transfer Learning Approach for Driver Drowsiness Detection Using Eye Movement Behavior

This paper presents a novel transfer learning technique to determine that someone is sleepy while driving by observing their eye movements. Current methods using physiological signals (i.e. heart rate or blood

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pressure) rely on having to touch the driver in some way and that can be very distracting for them. In this research, they used an established dataset comprising of 4,103 images of human eyes through the UnityEyes simulator, where images were taken using the simulated driver's eye both when open and closed providing balanced classes for each and resized to be 256x256 pixels with labels for if the eye is open/closed encoded as 0/1 respectively.

The methodology behind our proposed method for automatically determining whether a driver is drowsy is to first extract salient spatial features from images captured via VGG16 convolutional neural networks and then produce salient features with Light Gradient Boosting Machine (LGBM). Another component is an evaluation of four standard machine learning classifiers (K Nearest Neighbor, Light Gradient Boosting, Logistic Regression, Random Forest) and two standard Deep Learning models (Convolutional Neural Network and VGG16). Each model and hyperparameters were optimized and validated through the use of ten folds and hyperparameter tuning using ten-fold cross validation producing sufficient confidence when using this methodology. They also evaluate the computational complexity of each approach in terms of time.

The K-NC Algorithm achieved the highest accuracy (99%) and speed (0.00829 seconds) using VGLG features. Each competing algorithm had similar accuracy (98%) but a slower response time (RFC 0.0583 seconds, LGBM 0.0325 seconds; LR 0.00585 seconds). A few limitations should be considered, While the UnityEyes Dataset is a useful tool for creating synthetic eye features for training the models, it does not represent all of the different variables that can occur during real-world driving. Additionally, this study only focused on the eyes as an indicator for drowsiness. The lighting conditions of the environment can also affect feature extraction due to the variability of light sources within the vehicle; therefore, someone using video to detect drowsiness must carefully control their camera and light sources. Also, since cameras can have a wide variety of quality, then the quality of the image will impact both the accuracy of the model and its ability to detect drowsiness.

COMPARISON TABEL

No .	Year	Journal Name	Methodology	Evaluation Metrics	Results	Pros	Limitations
01	2024	IEEE Access	HOG + PCA feature extraction; 30-layer CNN; Hybrid HOG+PCA+CDLM; ML classifiers (SVM, DT, RF, XGBoost)	Accuracy, Precision, Recall, F1-measure	CDLM: 99.4% (ternary), 99.7% (binary), Hybrid: 99.6%, XGBoost: 97.4%	Comprehensive multimodule approach, illumination-invariant features, outperforms 12 methods.	Yawning-only focus, limited nighttime validation, 30-layer depth may cause degradation
02	2022	Applied Sciences	TCDCN for 68-point face detection, EAR/MAR/PERCLOS features, KNN classifier, weighted fatigue assessment model	Accuracy, EAR/MAR thresholds, PERCLOS	95.1% overall accuracy, 3-level fatigue classification	Multi-feature fusion, real-time capability, handles glasses occlusion	Simulated environment only, 30-min limited testing, night validation needed
03	2025	Applied Sciences	Dlib 68-point landmarks; Static threshold (ECR)	Accuracy, Sensitivity,	Static: 89.4% acc, 96.5% sens;	Training-independent; Cost-	Variable sensitivity (44-64%); Dlib

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			0.15, MAR 0.4); Adaptive frame threshold with time-based adjustment	Confusion Matrix	Adaptive: up to 98.2% acc, 64.3% sens	efficient; Long-haul focused	limitations at extreme angles; Linear approach limitations
04	2025	IEEE Access	Dlib HOG + Linear SVM; Adaptive threshold using 50 highest/lowest ECR; RF, DT, KNN, Multinomial classifiers	Accuracy, Sensitivity, Specificity, Precision, ROC-AUC	RF: 90.48%; DT: 89.44%; KNN: 88.63%; Blink detection: 99.78%	Individualized thresholds; Novel diagonal ECR computation; High blink accuracy	Eye-only focus; Impractical calibration phase; Limited subject diversity
05	2024	IEEE Access	Dlib 68-point landmarks; Shallow CNN (4 conv blocks); Transfer learning (VGG19, ResNet50, MobileNetV2, InceptionV3)	Accuracy, Precision, Recall, F1-score, ROC, PRC	Shallow CNN: 99.23% (D1), 99.14% (D2); Fast inference (1430ms)	Lightweight; Computationally efficient; Limited data requirement; Low inference time	Small/non-diverse datasets; No low-light validation; Eye-only focus
06	2023	IEEE Access	Cross-Guided Bilateral Filter; DTCWT + Walsh-Hadamard; Flamingo Search Algorithm; MCNN	Accuracy, Precision, Recall, F-measure, Kappa, FPR, FNR	YAWDD: 98.38%; NTHU-DDD: 98.26%; FPR: 0.013; FNR: <0.022	Advanced optimization; Low error rates; multi-dataset validation	Computational complexity; Cumulative pipeline errors; Limited demographic detail
07	2023	Journal of Imaging	Dlib HOG + MediaPipe face mesh; EAR/MAR/head pose; 15-frame window; RF, Sequential NN, SVM	Accuracy, Sensitivity, Specificity, Macro Precision, F1, ROC-AUC	RF: 99% acc, 0.999 AUC; NN: 96%; SVM: 80%; 33ms subsequent decisions	Multi-feature (eyes+mouth+head); Real-time; Systematic threshold selection	HOG fails in extreme conditions; Fixed window assumption; Simulated validation
08	2024	IEEE Access	VGLG (VGG16 + LGBM); KNC, LGBM, LR, RFC; 10-fold cross-validation	Accuracy, Precision, Recall, F1, Runtime complexity	KNC: 99% acc; Detection time: 0.0083s; 10-fold: 98% ± 0.003 -0.005	Ultra-fast detection; Transfer learning novelty; High cross-validation consistency	Synthetic dataset; Eye-only focus; Camera quality dependency